MASTER PROJECT KNOWLEDGE BASE

CoralGuard-X / Coral Reef Detection, Classification & Video Intelligence



Project evolution from the original box-only scope to the current accuracy-first architecture.

Document Version: 1.0
Reconstruction Date: 03 September 2026
Source of Truth: this project conversation and the project artifacts created/shared inside it

Document Purpose:

This document reconstructs and preserves the complete technical, operational, architectural, and troubleshooting knowledge accumulated throughout the project conversation.

> **Preservation Rule**
> Where the conversation contains superseded designs, this document preserves both the historical decision and the current replacement. Unknown or unverified details are explicitly marked Confirmed, Inferred, Unclear, Deprecated, or Pending.

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1. Project Overview and Reconstruction

Project name. CoralGuard-X is the latest name used for the integrated coral AI system. Earlier documentation also refers to the work generically as Coral Reef AI / Coral Detection & Classification.

Problem statement. The system must locate corals in underwater images/video, isolate each coral instance sufficiently well to reduce background contamination and overlap, classify the coral into the operational species labels available in the dataset, reject uncertain/non-coral candidates pragmatically, and produce stable annotated video suitable for research demonstration and eventual real-time use.

The scope evolved substantially. The original explicit requirement was Detection → Crop → Classification → Video → Real-Time, with segmentation explicitly excluded. After real video deployment exposed overlapping bounding boxes, background contamination, ambiguous colonies, and false labels on dead-looking/non-coral structures, instance segmentation was introduced as a practical refinement layer. The project then evolved through a Teacher→Student segmentation experiment and finally to an accuracy-first architecture that uses CoralSCOP directly as the primary mask refiner.

Requirement Type	Requirement / Constraint	Status
Core functional	Detect coral candidates in underwater images/video.	Confirmed
Core functional	Classify detected coral into six operational labels.	Confirmed
Core functional	Produce annotated video with stable IDs/species.	Confirmed
Quality	Handle small, crowded and overlapping colonies better than early baselines.	Confirmed
Quality	Reduce background/rock inside ROI before classification.	Confirmed
OOD/validity	Allow Unknown/Non-Coral rejection instead of forced six-class prediction.	Implemented pragmatically; not true open-set
Performance	Use local RTX A6000 and optimize long-running notebook stages.	Confirmed
Evaluation	Do not use held-out detector test for iterative tuning/model selection.	Confirmed
Taxonomy	Do not invent biological certainty where labels are ambiguous/typo-prone.	Confirmed
Deployment	Near-term requirement: generate a presentable first video quickly for academic review.	Current priority

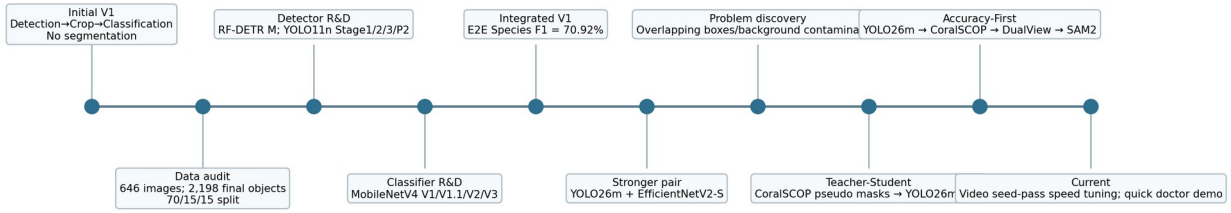
Target users / stakeholders. No formal product user model was implemented. The conversation identifies the user as the research/development operator and an academic “doctor” as a review/demo stakeholder. There is no implemented web application, customer portal, or enterprise user management layer.

Operational class labels (as audited in the annotation set):

- Euphflfiaancora
- Favosites
- Platygyra
- Sarcophyton
- Sinularia
- WavingHand

Taxonomy caution
Platygyra, Sarcophyton and Sinularia are genus-level names. Euphflfiaancora may correspond to Fimbriaphyllia ancora; Favosites may be a typo for Favites; WavingHand is a common-name style label. The conversation explicitly required not overstating taxonomic certainty.

2. Project Timeline



Logical project timeline. Dates are not assigned where the conversation did not provide an explicit date.

ID	Milestone	What changed	Status
T01	Initial V1 scope	Detection → bounding box → ROI crop → species classification → annotated video → real-time. Segmentation explicitly excluded.	Deprecated scope constraint
T02	Dataset audit and canonicalization	646 image/JSON pairs audited; one duplicate object removed; EXIF orientation normalized; leakage-safe split established.	Final data foundation
T03	RF-DETR Medium baseline	10-epoch detector reached strong validation performance; demonstrated that the dataset could support much stronger localization than tiny YOLO baseline.	Historical benchmark
T04	YOLO11n baseline + Stage 2	YOLO11n 10-epoch baseline followed by 832px underwater-focused fine-tuning.	Superseded detector path
T05	YOLO11n Stage 3	Targeted small/crowded sliced fine-tuning at 960px; became the lightweight detection baseline. SAHI inference later failed to help.	Historical baseline
T06	YOLO11n P2 Stage 4	P2/P3/P4/P5 experiment tried to improve small objects; transfer mismatch and short training left it under-converged.	Not adopted
T07	MobileNetV4 classifier evolution	V1 → V1.1 → V2 Sarcophyton-targeted → V3 288px +10% context + hard mining.	Historical classifier baseline
T08	Integrated Stage3 + MobileNetV4 V3	Validation E2E Species F1 = 70.92%; failure attribution showed detector/localization dominated errors.	Important benchmark
T09	Video stabilization	BoT-SORT, bbox EMA, gap hold, classifier probability EMA; FFmpeg H.264 replaced low-quality mp4v.	Retained concepts
T10	Stronger model pair	YOLO26m + EfficientNetV2-S became the stronger deployment pair; local Windows/A6000 migration followed.	Pre-segmentation baseline
T11	Real-video failure discovery	Overlapping boxes, background inside ROIs, Unknown/incorrect predictions, dead-looking/algae-like structures exposed limitations of box-only crop classification.	Architecture pivot trigger
T12	CoralGuard-X hybrid	YOLO26m candidate detection + CoralSCOP box-prompted masks + classifier + SAM2 proposed.	Superseded by Teacher→Student then accuracy-first
T13	Teacher→Student V2	CoralSCOP generated pseudo instance masks; YOLO26m-SEG student trained for 150 epochs.	Completed experiment
T14	Student evaluation	Pseudo-mask mAP50:95 0.5905; original-GT box operating F1 0.7311. Student segmentation useful, detector behavior weaker than old YOLO26m.	Decision point
T15	Final accuracy-first architecture	Old YOLO26m kept as primary detector; CoralSCOP used directly on detector candidates; Dual-View classifier + OOD gate + SAM2.1.	Current final architecture
T16	Runtime optimization	Dual-view data build, classifier training, and video seed pass were optimized. Refresh stride moved from	Current work

		1 to 5 for speed; quick one-video demo path requested.	

3. Full Architecture Documentation

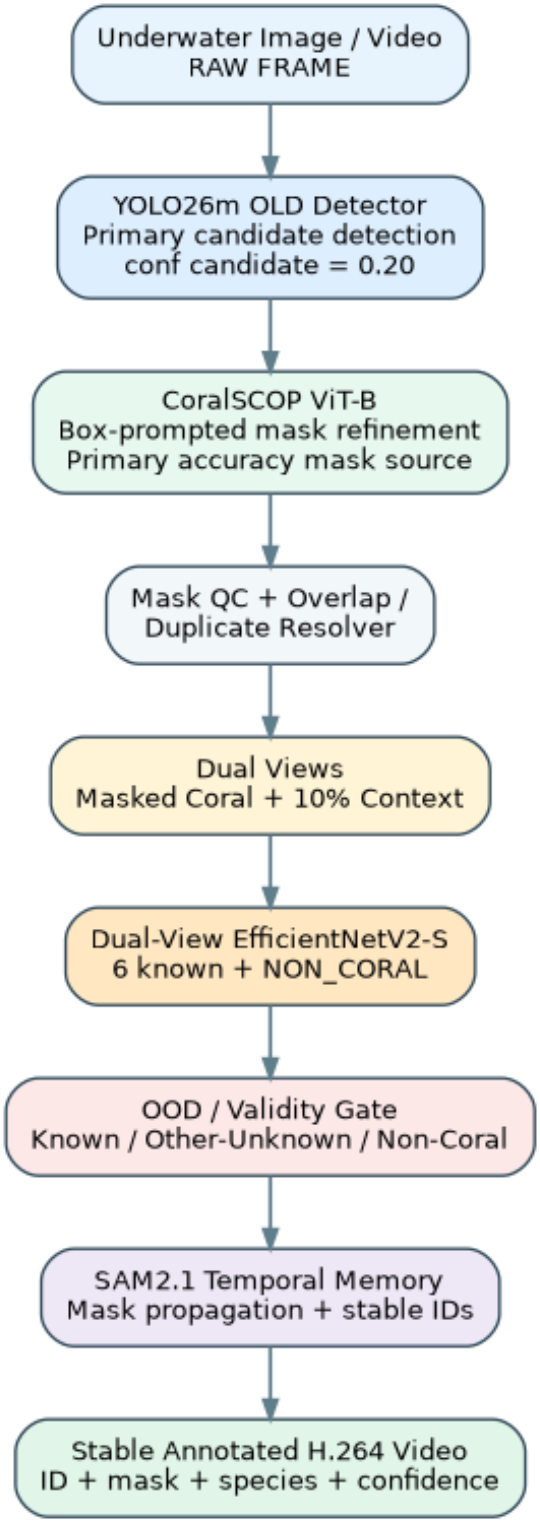


Figure 3-1. Final accuracy-first system architecture.

This is a research/ML notebook system rather than a conventional web application. Therefore “frontend”, “backend”, “database”, and “permissions” are mapped to their functional equivalents below rather than invented as nonexistent services.

Component	Purpose	Inputs	Outputs	Dependencies
Notebook Control Layer	VS Code Jupyter notebooks on Windows; exposes config cells, training/evaluation cells, progress output and artifact paths.	Config variables, file paths, checkpoints	Models, manifests, reports, videos	Python/PyTorch/Ultralytics/ timm/OpenCV
File-System Data Layer	Local workspace stores datasets, masks, manifests, model checkpoints, caches and outputs.	COCO/CSV/JPEG/PNG/ JSON/ZIP	Derived datasets/checkpoints	D:\Coral_Local_Workspace
YOLO26m Detector	Primary candidate detector in current architecture.	RGB image/frame	Candidate bounding boxes + confidence	Ultralytics 8.4.117; old verified checkpoint
CoralSCOP	Accuracy-first instance mask refiner using YOLO boxes as prompts.	RGB frame + box prompt	Precise binary mask + score	CoralSCOP ViT-B checkpoint, segment_anything-style API
Mask QC / Duplicate Resolver	Chooses/filters masks and suppresses duplicate overlapping hypotheses.	Masks, detector boxes, scores	Clean instance list	IoU, containment, mask quality logic
Dual-View EfficientNetV2-S	Shared backbone classifier over masked coral and +10% context view.	Two 300×300 views	7-way logits/embedding	Old EfficientNetV2-S transfer weights + new fusion/head
OOD / Validity Gate	Prevents forced known-species assignment when confidence/margin/embedding is weak or NON_CORAL is strong.	Probabilities + embedding	Known / Other-Unknown / Non-Coral	Validation-derived prototypes/thresholds
SAM2.1 Temporal Memory	Propagates masks and stabilizes object identity between refresh frames.	Seed masks + video frames	Per-frame masks + IDs	facebook/sam2.1-hiera-large
FFmpeg Output Layer	Creates high-quality H.264 MP4, optionally preserving audio.	Annotated raw frames + source audio	Final MP4	libx264; CRF/preset configurable

Current inference sequence:

```
RAW FRAME
  → YOLO26m old detector (candidate boxes)
  → CoralSCOP on each candidate box
  → mask quality + duplicate/overlap resolution
  → masked coral view + 10% context view
  → Dual-View EfficientNetV2-S
  → known / Other-Unknown / Non-Coral gate
  → SAM2.1 temporal memory and stable IDs
  → H.264 annotated video
```

Deprecated / alternate architecture: YOLO26m-SEG Student as primary segmenter. This remains a useful fast/backup model but is no longer the accuracy-first primary spatial path because its original-GT detection operating point was weaker than the old YOLO26m detector.

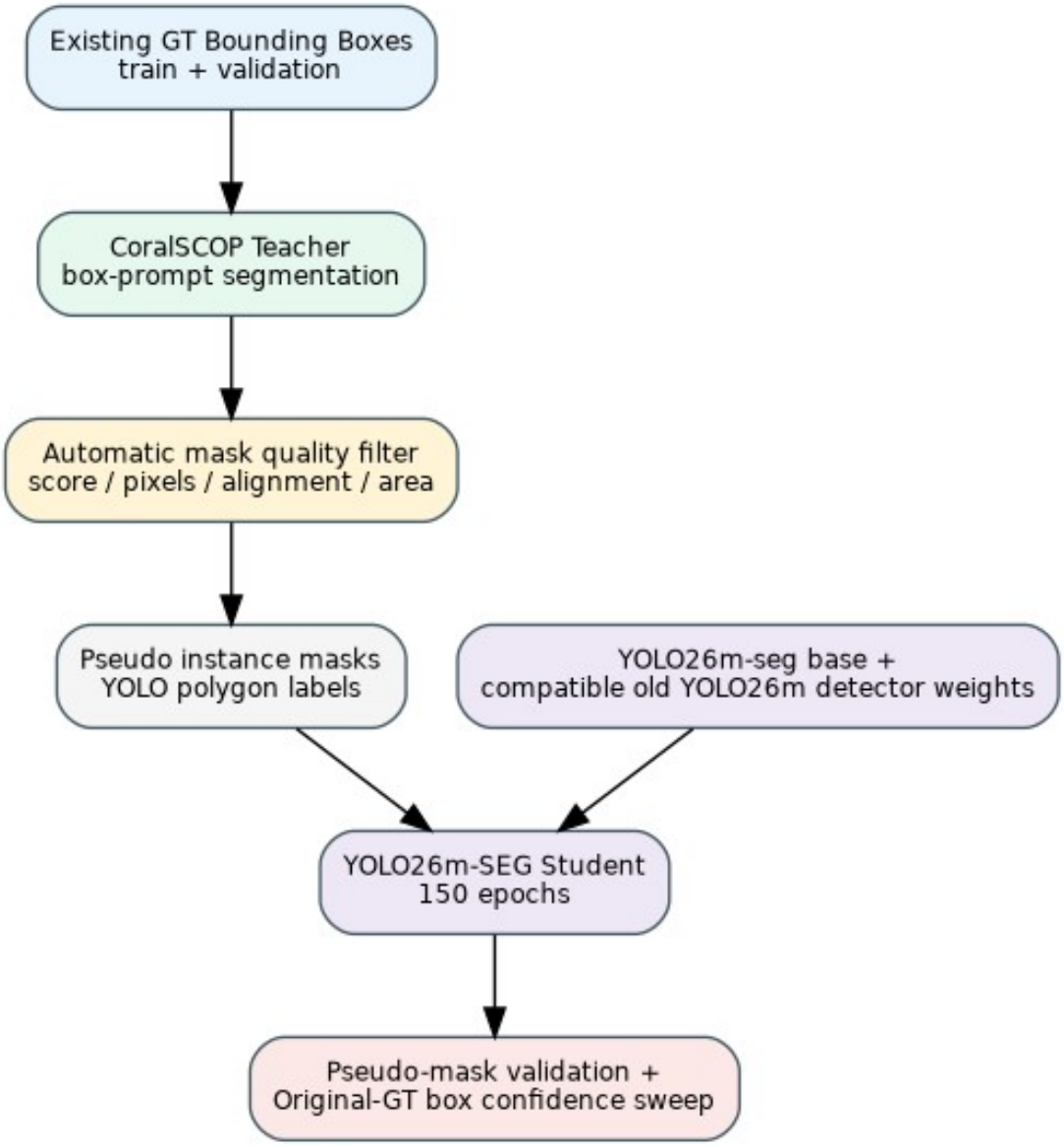


Figure 3-2. Teacher→Student training architecture retained as a completed experiment and possible fast-mode model.

Mermaid representation of the current architecture:

```
flowchart TD
    A[RAW underwater frame] --> B[YOLO26m old detector]
    B --> C[Candidate boxes]
    C --> D[CoralSCOP box-prompt masks]
    D --> E[Mask QC + overlap/duplicate resolver]
    E --> F1[Masked coral view]
    E --> F2[+10% context view]
    F1 --> G[Dual-View EfficientNetV2-S]
    F2 --> G
    G --> H[OOD / validity gate]
    H --> I[SAM2.1 temporal memory]
    I --> J[Stable annotated H.264 video]
```

4. Data Architecture

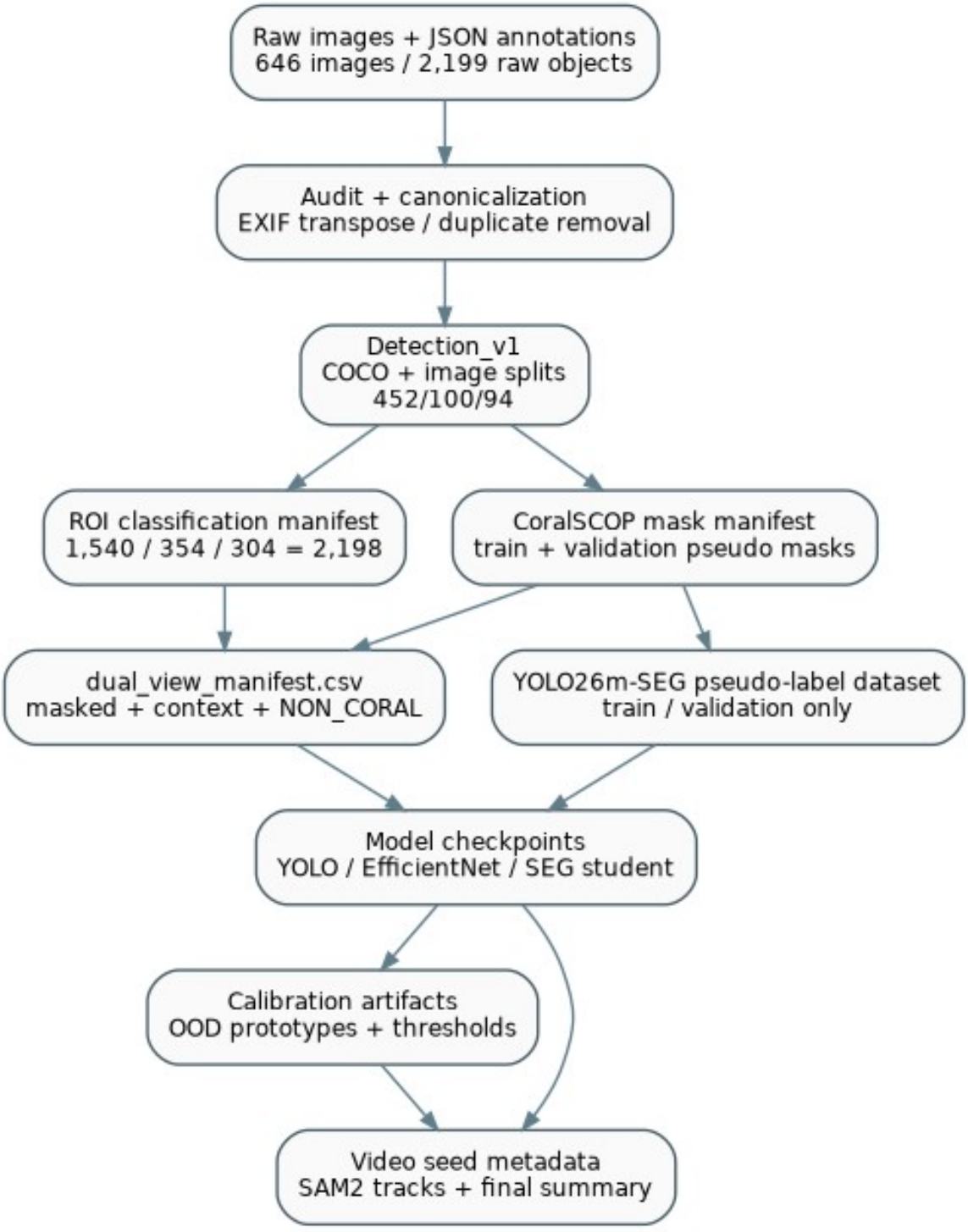


Figure 4-1. Dataset and artifact data flow.

The project uses a file-based data architecture: COCO JSON, CSV manifests, image folders, PNG mask files, YAML dataset configs, PyTorch checkpoints and JSON/CSV reports. No relational database was implemented.

4.1 Canonical dataset inventory

Dataset layer	Train	Validation	Test	Total / Notes
Detection images	452	100	94	646 images
Detection boxes	1,540	354	304	2,198 final objects

Classification ROIs	1,540	354	304	2,198 ROIs; same split lineage
CoralSCOP teacher masks	train generated	validation generated	NOT generated for tuning	Pseudo masks, not human GT
Seg student validation	-	100 images / 352 pseudo-mask instances	test untouched	2 validation objects filtered/absent from pseudo-mask dataset

Raw audit history: 2,199 raw objects were found; one duplicate annotation was removed, leaving 2,198 final ground-truth objects. Forty images required EXIF orientation normalization during canonicalization.

4.2 Core entities / files and Data Dictionary

COCO Detection JSON

Field	Type	Purpose	Source	Validation / Relationships
images[].id	integer (observed)	Image identifier	Dataset audit/canonicalization	Unique within split; referenced by annotations
images[].file_name	string (observed)	Image filename/path	Canonical image folder	Must resolve to an image file
annotations[].id	integer (observed)	Annotation identifier	Annotation conversion	Used to name/associate pseudo masks
annotations[].image_id	integer (observed)	Parent image reference	COCO JSON	FK-like reference to images.id
annotations[].category_id	integer (observed)	Detector category	Canonicalization	Single detector class: target_coral
annotations[].bbox	[x,y,w,h] numeric	Ground-truth rectangle	Raw annotation conversion	Converted to xyxy for many internal calculations

classification_roi_manifest_split.csv

Field	Type	Purpose	Source	Validation / Relationships
roi_id	string	Unique ROI record	ROI dataset builder	Used to name context/masked images
image_stem	string	Source image key	Source image filename	Used for CoralSCOP mask mapping
raw_detection_label	string	Operational species label	Original annotations	One of six audited labels
x1_px/y1_px/x2_px/y2_px	numeric	ROI box in pixels	Canonical annotation	Used for crop and mask matching
split	string	train/validation/test	Leakage-safe split	Must preserve source-image split

coralscop_mask_manifest.csv

Field	Type	Purpose	Source	Validation / Relationships
split	string	Data split	Mask generation loop	train/validation only during tuning
ann_id	integer	Source annotation ID	COCO annotation	Links pseudo mask to prompt object
image_id	integer	Source image ID	COCO image	Parent reference
image_path / image_stem	string	Source image locator/key	Canonical dataset	Used by dual-view mapping
mask_path	string	PNG mask location	CoralSCOP generation	Must exist and decode to 2D boolean mask
mask_score	float	Teacher mask score	CoralSCOP predictor	Minimum quality gate typically 0.35
area/alignment metrics	float where present	Pseudo-mask audit	mask quality logic	Teacher agreement indicators, not independent GT accuracy

dual_view_manifest.csv

Field	Type	Purpose	Source	Validation / Relationships
split	string	train/validation	ROI split	No detector test used for training
label	string	6 species + NON_CORAL	ROI label or generated negative	Classifier target
context_path	string	+10% context JPEG	Dual-view builder	JPEG quality 95
masked_path	string	Foreground-focused JPEG	CoralSCOP mask/proxy mask	JPEG quality 95

source	string	Provenance marker	Builder	e.g., CORALSCOP_CORAL / BACKGROUND_NEGATIVE

student_operating_config.json

Field	Type	Purpose	Source	Validation / Relationships
best_seg_conf_original_gt	float	Best validation operating confidence	Original-GT confidence sweep	Observed result 0.40
model/checkpoint refs	string	Student checkpoint metadata	Cell 9	Promoted to models/best_yolo26m_seg_student.pt

ood_calibration.json + ood_prototypes.npz

Field	Type	Purpose	Source	Validation / Relationships
known_classes	list	Known species classes	Classifier metadata	Six known species
similarity_thresholds	mapping	Per-class cosine threshold	Known validation embeddings	3rd percentile scheme
known_conf_threshold	float	Minimum known-class confidence	Calibration config	0.50 in current design
known_margin_threshold	float	Top1-top2 margin gate	Calibration config	0.08
non_coral_threshold	float	NON_CORAL rejection gate	Calibration config	0.45
proto__<class>	vector	Class embedding prototype	Train embeddings	Stored in NPZ

video seed_metadata.json / summary.json

Field	Type	Purpose	Source	Validation / Relationships
track_id	integer	Temporal object ID	Refresh-frame association	May fragment if association is weak
frame_idx	integer	Conditioning frame index	Video seed pass	Depends on refresh_stride
mask_path	string	Saved conditioning mask	CoralSCOP seed pass	PNG; compression optimized during speed work
box	numeric list	Object rectangle	Mask bbox	Used for association/overlay
det_conf	float	YOLO detection confidence	Detector	Candidate threshold currently 0.20
label / cls_conf	string/float	Classifier result at seed	Dual-view classifier	May be smoothed later
refresh_stride	integer	Keyframe cadence	Runtime config	Notebook default 3; experiments 1 and 5; quick-demo tuning pending
output	string	Final video path	FFmpeg render	Accuracy-first final MP4



Figure 4-2. ERD-style relationship model for the file-based project entities.

Data provenance rule
Mask mAP against CoralSCOP pseudo masks measures agreement with the teacher, not independent segmentation truth. The detector held-out test remains reserved for final selection/evaluation.

5. Complete Workflow Extraction

Workflow	Trigger / Input	Processing & Validation	Output
WF-01 Dataset Audit & Canonicalization	Raw image + JSON folders	Inventory pairs; validate annotations; remove duplicate; EXIF transpose; convert to canonical RGB JPEG; split by image 70/15/15; write COCO/split artifacts.	646 valid pairs, 2,198 final GT boxes. Test split isolated.
WF-02 Detector Training	Canonical Detection_v1	Train baseline or fine-tune chosen YOLO/RF-DETR model on train; validate only on validation; checkpoint best weights; tune confidence on validation.	Detector checkpoint + metrics + confidence operating point.
WF-03 Classification ROI Training	ROI manifest + GT boxes	Crop/pad ROI; preserve split lineage; train classifier; hard-mine/target confusions without touching test for	Classifier checkpoint + class metrics.

		model selection.	
WF-04 CoralSCOP Pseudo-Mask Generation	Train/validation images + GT boxes	Set image once; prompt each GT box; choose best mask; normalize to 2D; quality filter; save PNG + manifest.	Teacher pseudo masks.
WF-05 YOLO26m-SEG Student	Pseudo masks + YOLO26m-seg + old detector weights	Convert masks to YOLO polygons; train 150 epochs; validate pseudo masks; sweep confidence against original GT boxes.	Student segmenter; currently backup/fast mode.
WF-06 Dual-View Dataset	ROI manifest + CoralSCOP masks	Map ROI↔mask by image key + box IoU; save +10% context and masked foreground; create conservative NON_CORAL negatives.	dual_view_manifest.csv.
WF-07 Dual-View Classifier	Dual-view pairs + old EfficientNetV2-S weights	Shared EfficientNetV2-S backbone; fuse context/masked embeddings; train up to 150 epochs; select by validation Macro-F1.	Best dual-view checkpoint.
WF-08 OOD Calibration	Train/validation embeddings	Compute normalized embeddings; class prototypes; per-class similarity thresholds; set confidence/margin/NON_CORAL gates.	OOD calibration artifacts.
WF-09 Accuracy-First Image Inference	Image/frame	YOLO candidates → CoralSCOP each box → mask QC/duplicate resolver → dual-view classifier → OOD gate.	Instance masks + species/validity labels.
WF-10 Video Seed Pass	Video + accuracy-first spatial stack	Extract frames; process refresh frames; associate tracks; save CoralSCOP masks as SAM2 conditioning records.	seed_metadata.json and seed masks.
WF-11 SAM2 Final Video	Frame folder + seed masks + classifier	Load SAM2.1; add mask prompts; propagate; classify periodically; EMA probabilities; render overlays; encode H.264.	Final MP4 + JSON/CSV summary.

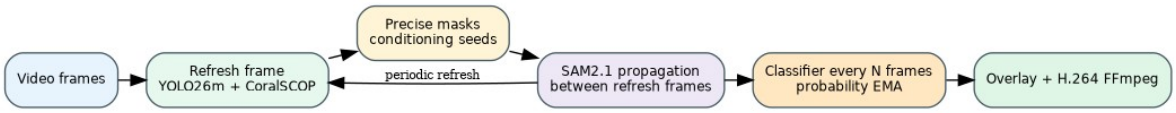


Figure 5-1. Final video runtime workflow.

5.1 Error scenarios common to workflows

- Missing checkpoint/path: fail fast with FileNotFoundError rather than silently downloading/retraining the wrong model.
- Mask dimensionality mismatch: normalize/squeeze to a strict 2D boolean mask before indexing RGB arrays.
- Mask/image resolution mismatch: nearest-neighbor resize mask to source image dimensions.
- Pseudo mask missing/failed: record failure reason and exclude rather than invent a polygon.
- Validation/test leakage risk: test must not participate in threshold tuning or pseudo-label model selection.
- Unknown taxonomy/background: use pragmatic rejection label rather than assigning a known species with unsupported certainty.
- Video tracking fragmentation: currently observed as many track IDs; needs stronger association/pruning before final runtime benchmark.

6. Business Rules / Technical Governance Rules

Rule ID	Rule	Where Applied	Reason	Current Status
BR-001	Detector is single-class target_coral; species classification occurs downstream.	Detection dataset/model	Separates localization from taxonomy	Final
BR-002	Split by source image; expected detection split = 452/100/94 images.	All model training	Prevent leakage	Final
BR-003	Do not use held-out detector test during iterative training/model selection.	Detector/segmentation	Preserve independent evaluation	Final
BR-004	Old classifier test was historically exposed; do not reuse it for future model selection.	Classification	Avoid optimistic bias	Final caution
BR-005	Use +10% context around coral crop for classification.	Classifier	V3 improved robustness while retaining morphology	Final
BR-006	Avoid blur/downscale/heavy JPEG/noise for texture-sensitive classifier training.	Classifier augmentation	Coral morphology/texture are discriminative	Final
BR-007	CoralSCOP pseudo masks	Segmentation evaluation	Scientific validity	Final

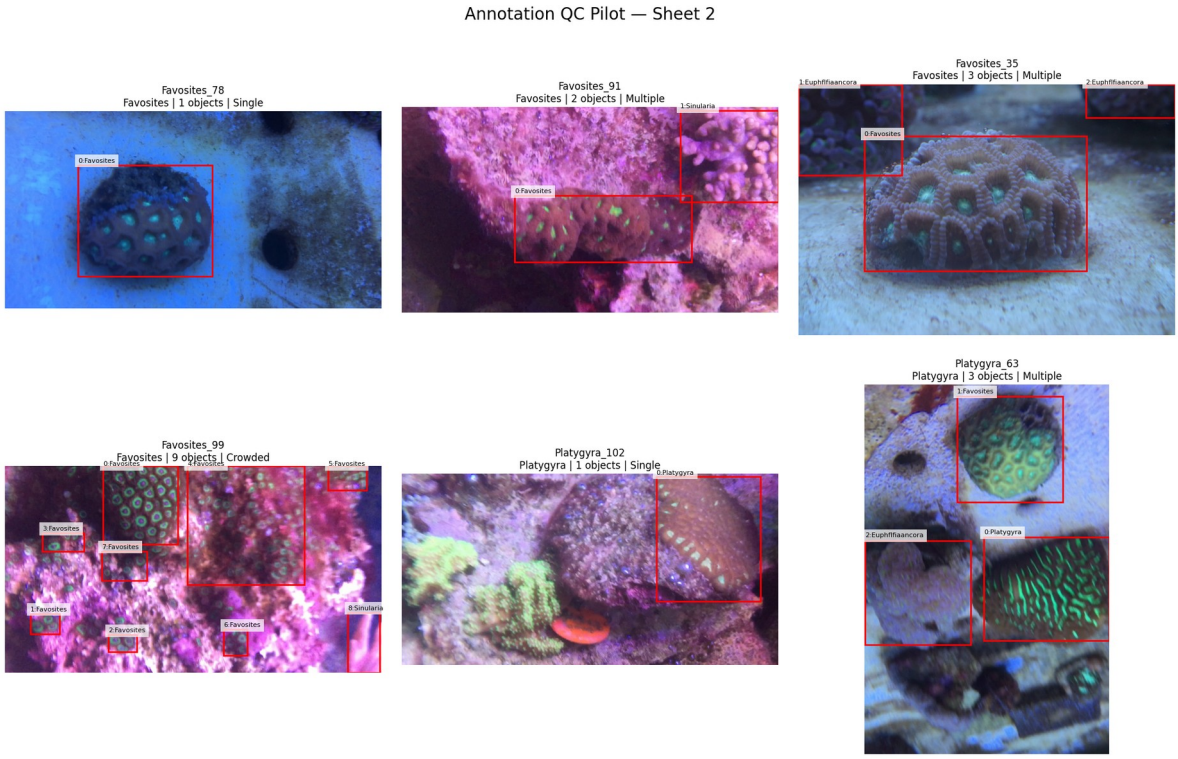
	are teacher labels, not human GT.			
BR-008	No explicit Dead Coral or Algae class exists; do not invent these labels.	Classifier/output	Annotation truth constraint	Final
BR-009	Unknown is pragmatic rejection, not proven open-set recognition.	OOD gate	No explicit unseen-species validation set	Final
BR-010	Candidate detection confidence in accuracy-first runtime is intentionally low (0.20).	Video/image inference	Favor recall; downstream mask/classifier can reject	Current
BR-011	CoralSCOP mask score minimum typically 0.35; minimum mask pixels 40.	Mask QC	Reject empty/weak masks	Current
BR-012	Duplicate masks suppressed around mask IoU 0.65 / containment 0.85 with box-overlap support.	Duplicate resolver	Avoid multiple labels on same colony	Current
BR-013	SAM2 classification is periodic and probabilities are EMA-smoothed (alpha 0.25).	Video	Reduce label flicker	Current
BR-014	Final H.264 quality mode used CRF 12 / slow; quick-demo mode may use CRF 16 / veryfast.	Video export	Trade quality vs urgent demo speed	Current/temporary
BR-015	REFRESH_STRIDE is a performance/accuracy lever. Notebook default 3; user tested 1 then 5; faster demo settings are allowed.	Video seed pass	CoralSCOP is the dominant refresh cost	Current runtime tuning

7. Problems & Troubleshooting Knowledge Base

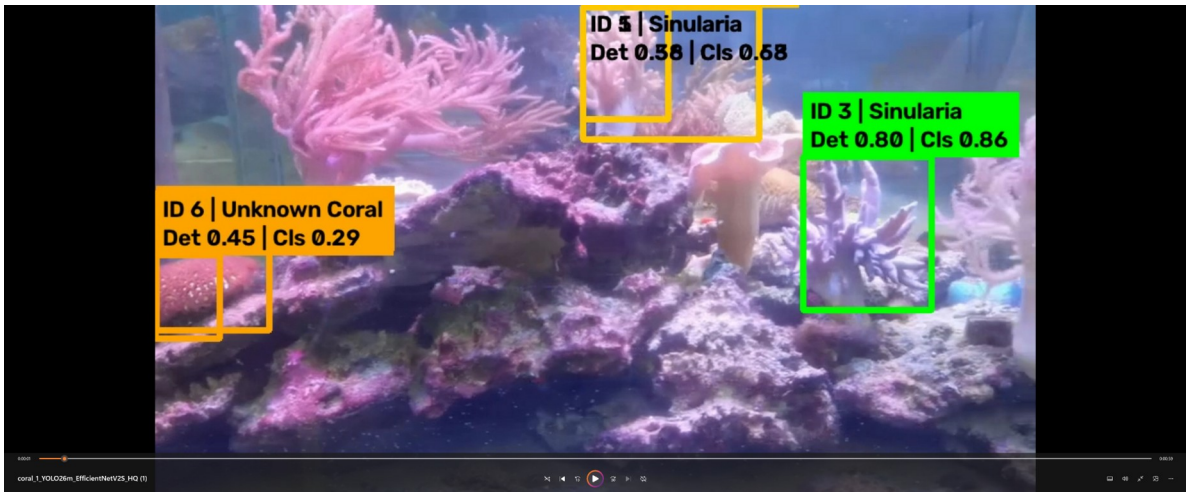
Issue ID	Component	Problem / Symptoms	Root Cause	Final Fix	Lesson Learned	Status
ISS-001	Data audit	One duplicated object annotation existed.	Object count mismatch / duplicate GT.	Duplicate detected during audit; removed.	Canonical final object count fixed at 2,198.	Resolved
ISS-002	Image orientation	40 images changed orientation under EXIF transpose.	Raw file pixels did not reflect display orientation consistently.	Canonical RGB images generated after EXIF transpose.	Use EXIF-aware loading before annotation geometry is finalized.	Resolved
ISS-003	Taxonomy	Several labels are typo/common-name/uncertain taxonomy.	Dataset operational labels are not all clean scientific names.	Preserve original label vocabulary; document candidate taxonomic corrections only.	Do not silently rename biological classes.	Open documentation issue
ISS-004	YOLO11 Stage2	Small and crowded recall remained weak.	P3-only small-object limitations plus scene crowding/localization.	Targeted small/crowded sliced training at 960px.	Stage3 improved small/crowded metrics.	Resolved/benchmark
ISS-005	Stage3 slicing	Original positive-slice selection did not guarantee small GT inclusion.	Random positive slices could miss the target small object.	Stage4 fixed selection logic to prioritize actual small objects.	Selection semantics matter as much as slice count.	Resolved in later experiment
ISS-006	SAHI	SAHI inference performed badly despite the small-object motivation.	Sliced inference/merge behavior did not fit this dataset/model setup.	Evaluated then rejected.	Do not keep SAHI in final runtime just because it is theoretically relevant.	Rejected
ISS-007	YOLO11 P2 Stage4	P2 model severely underperformed Stage3 after 15 epochs.	Architecture/head changed and only partial transfer occurred; run was under-converged.	Do not interpret as proof P2 is inherently bad; experiment not promoted.	Architecture changes may need longer recovery training.	Not adopted
ISS-008	Classifier	Sarcophyton was a weaker class.	Class-specific confusion/visual similarity.	Targeted V2 fine-tune, then V3 288px + context + hard mining.	Class-targeted hard mining can outperform generic augmentation.	Resolved to V3 baseline
ISS-009	Video V1	Boxes/species flickered frame-to-frame.	Independent frame predictions and unstable detector boxes/probabilities.	BoT-SORT + bbox EMA + gap hold + probability EMA.	Temporal smoothing mitigates but does not mathematically eliminate flicker.	Resolved conceptually
ISS-010	Video quality	OpenCV mp4v output looked visibly degraded.	Codec/encoding quality.	Switch to FFmpeg H.264, high-quality CRF 10/12.	Encoding layer matters independently of model accuracy.	Resolved
ISS-011	Closed-set classifier	Non-coral/unseen structures were forced into known labels.	Six-class softmax has no native unknown class.	Confidence/margin rejection, later NON_CORAL training + embedding OOD gate.	Unknown remains pragmatic without an explicit unknown benchmark.	Partially resolved
ISS-012	Deployment	Bounding boxes included rock/background and overlapped colonies.	Box-only localization cannot isolate irregular adjacent coral shapes.	Architecture pivot to instance segmentation via CoralSCOP.	Real-video failure modes should drive architecture, not only validation mAP.	Resolved architecturally
ISS-013	Initial CoralGuard	CoralSCOP only	User wanted the	Teacher→Student	Clarify whether	Superseded

	hybrid	refined masks at inference; YOLO itself was not retrained for masks.	segmenter to learn from teacher masks, not only call teacher at runtime.	pseudo-label architecture created.	"teacher-student" means pseudo-label supervision vs logit distillation.	
ISS-014	Cell 5	defaultdict missing import.	Notebook cell dependency oversight.	Add from collections import defaultdict.	Standalone/restart-safe cells must own their imports.	Resolved
ISS-015	Cell 5	coralscop_predictor undefined after restart/out-of-order execution.	Runtime state from Cell 4 was lost.	Run CoralSCOP load cell / use restart recovery cell.	Notebook state must be recoverable from disk artifacts.	Resolved
ISS-016	Mask helper	mask_bbox raised "too many values to unpack".	Stored/read mask sometimes had singleton/channel dimension.	normalize_mask_2d using squeeze and channel handling before np.where.	Never assume PNG mask loads as H×W.	Resolved
ISS-017	Student dataset Cell 6	KeyError: score in mask audit.	Empty-mask branch returned incomplete dictionary.	mask_quality always returns score/area_ratio/bbox_iou/pixels/reason.	All error branches must honor the same schema.	Resolved
ISS-018	Student evaluation	YOLO26m-SEG was good on pseudo masks but weaker as detector on original GT.	Training objective optimized teacher-mask agreement, not necessarily old box operating behavior.	Keep old YOLO26m as primary detector; use CoralSCOP directly for accuracy path; SEG student becomes backup/fast mode.	Evaluate each component on the ground truth relevant to its intended job.	Final decision
ISS-019	Windows restart	Variables/models disappeared after device restart.	Notebook in-memory state is ephemeral.	Recovery cell reloads checkpoints, configs, mask manifest, helper functions.	Persist every important artifact and make restarts first-class.	Resolved
ISS-020	Dual-view Cell 7	IndexError boolean index did not match RGB channel axis.	local_mask had shape H×W×1 while context was H×W×3.	Robust mask normalization/squeeze; resize nearest-neighbor if needed; index with strict H×W bool mask.	Shape assertions belong immediately before boolean image indexing.	Resolved
ISS-021	Dual-view build speed	Building 1,894 ROI pairs was slow.	Repeatedly opening same source image and JPEG I/O.	Turbo grouped processing by source image; resume skip; fast OpenCV read; high-quality JPEG retained.	Group work by shared expensive input.	Resolved
ISS-022	Classifier training speed	Dual-view EfficientNetV2-S was very slow.	Two image views, repeated JPEG decode/resize, two separate backbone forwards.	RAM cache resized images; one concatenated backbone forward; AMP; channels_last; TF32; fused AdamW; resume checkpoints.	Optimize input pipeline and GPU utilization before reducing model/image quality.	Optimization supplied; final speed not yet reported
ISS-023	Jupyter warning	TqdmWarning: IProgress not found.	ipywidgets notebook progress integration missing.	Ignored as non-functional warning.	Do not treat UI progress warning as training failure.	Benign
ISS-024	Video seed pass speed	REFRESH_STRIDE=1 caused ~1.5–2.1 s per refresh and very long runs.	YOLO + CoralSCOP were executed on every frame.	Increase REFRESH_STRIDE to 5; CoralSCOP only on refresh frames and SAM2 handles temporal propagation.	A "seconds/frame" log can actually mean seconds/refresh; interpret denominator correctly.	Improved but still being tuned
ISS-025	Video track count	Track IDs grew to ~67–69 in sample seed pass.	Association fragmentation / new IDs accumulating.	Track-pruning/ stronger association identified as next fix; quick demo proposed top 25 stable tracks.	Uncontrolled ID growth can make SAM2 the next bottleneck.	Pending
ISS-026	Urgent demo	Need first video for doctor quickly rather than all five.	Batch pipeline prioritized completeness over near-term demo.	Process first completed seed metadata only; keep top 25 stable tracks; classifier stride 5; FFmpeg veryfast CRF16.	Production workflow should support demo/fast mode separately.	Temporary demo mode

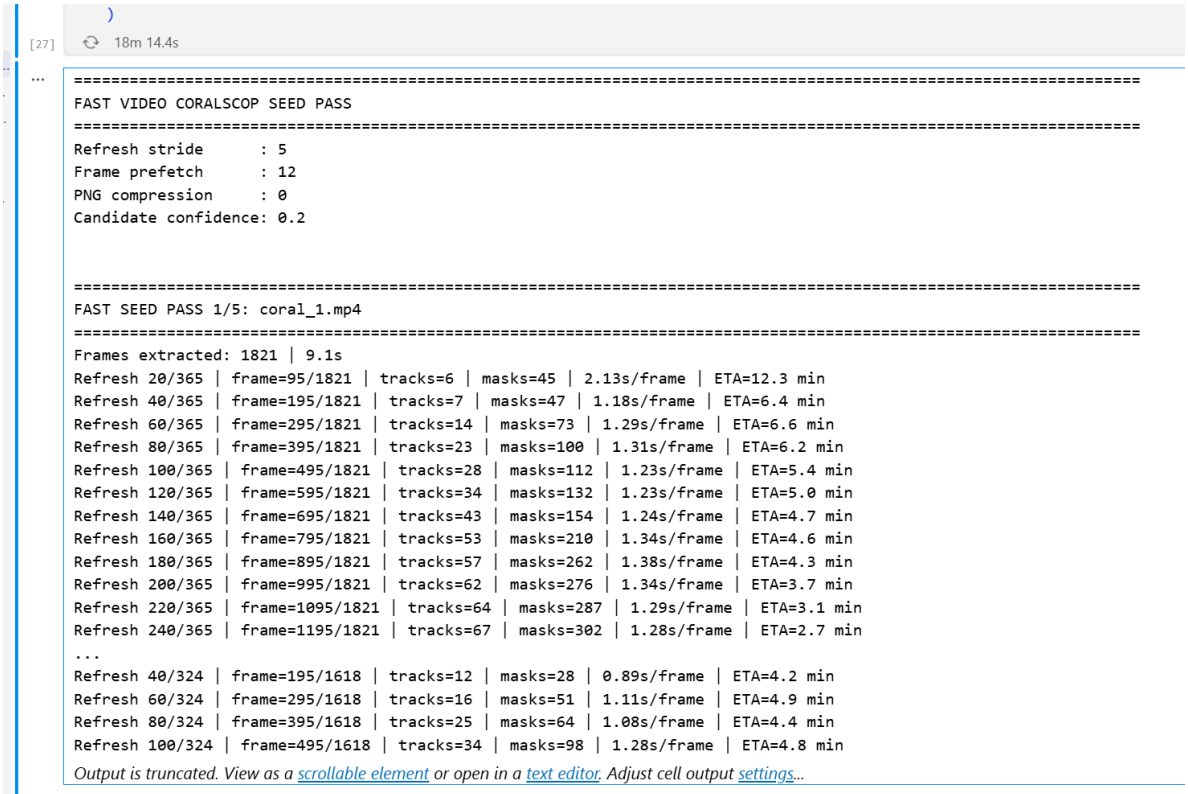
7.1 Screenshots preserved from troubleshooting



Annotation QC example showing single, multiple and crowded coral boxes. This context motivated explicit small/crowded analysis.



Real-video deployment screenshot showing overlapping detections and an “Unknown Coral” prediction. This was a key trigger for the segmentation pivot.



Current video seed-pass tuning with REFRESH_STRIDE=5. The output shows fewer refresh operations but continued CoralSCOP cost and rapid growth in track IDs.

8. Decisions Log

Decision ID	Decision	Alternatives	Reason	Consequence	Status
DEC-001	Start with detection + ROI classification; no segmentation.	Segmentation-first pipeline	Simpler original scope and available box annotations.	Built strong box/classification baseline.	Deprecated
DEC-002	Single-class detector target_coral, species classifier downstream.	6-class detector	Separates localization and species learning; supports dedicated classifier.	Two-stage pipeline.	Final
DEC-003	Use validation for threshold tuning; preserve detector test.	Tune on test	Scientific evaluation hygiene.	Independent test remains available.	Final
DEC-004	Stage3 YOLO11n as lightweight baseline rather than SAHI/P2.	SAHI / P2 Stage4	Measured results and deployment simplicity.	Stage3 retained for historical comparison.	Historical final for YOLO11 path
DEC-005	MobileNetV4 V3 as lightweight classifier baseline.	Earlier V1/V1.1/V2	V3 best test/validation balance with context/hard mining.	Historical classifier benchmark.	Historical
DEC-006	Move to YOLO26m + EfficientNetV2-S for stronger pair.	Keep only tiny models / RF-DETR deployment	Stronger practical capacity and integrated local pipeline.	New pre-seg baseline.	Superseded by final spatial design but detector retained
DEC-007	Introduce instance segmentation after real-video evidence.	Continue box-only tuning	Boxes could not isolate irregular overlapping corals.	Scope expanded beyond original “no segmentation” rule.	Final architectural pivot
DEC-008	Use CoralSCOP as domain-specific mask teacher.	Generic semantic segmentation	Box-prompted coral-specific segmentation useful without manual mask GT.	Pseudo masks and direct runtime masks.	Final
DEC-009	Train YOLO26m-SEG Student from CoralSCOP pseudo masks.	Teacher-only inference	Wanted a learned fast instance segmenter.	Completed 150-epoch student experiment.	Completed but not primary runtime
DEC-010	Do not promote SEG student as primary detector.	Student-only detector/segmenter	Original-GT operating F1 weaker than old YOLO26m.	Old YOLO26m remains candidate detector.	Final
DEC-011	Accuracy-first path uses CoralSCOP directly on YOLO candidates.	Student mask primary + CoralSCOP fallback	Accuracy prioritized over raw speed; teacher likely stronger than imitation student.	CoralSCOP becomes primary mask source.	Current final
DEC-012	Use SAM2.1 for temporal memory instead of running full spatial stack every	Independent-frame inference only	Reduce flicker/cost and propagate precise masks.	Keyframe/refresh architecture.	Final

	frame.				
DEC-013	Train Dual-View classifier with masked + context images and NON_CORAL proxy negatives.	Single crop only	Preserve species texture/context while giving clean foreground view and validity signal.	7-way classifier + OOD gate.	Current
DEC-014	Keep segmentation student as Fast/Backup mode.	Discard student	150-epoch work remains valuable for speed/recovery.	Potential quality vs speed operating modes.	Final secondary
DEC-015	Runtime tuning uses refresh stride rather than raising detector confidence.	Raise DET_CANDIDATE_CONF	Higher confidence risks missing small corals; stride directly attacks CoralSCOP cost.	DET candidate remains 0.20 during tuning.	Current

9. Code Knowledge Extraction

Current / final code is distributed primarily across two notebooks: the completed Teacher→Student notebook and the later Accuracy-First notebook. Several fixes and speed patches were supplied after the latest notebook was generated; those patches are documented here because they are not guaranteed to be integrated into the saved notebook file.

9.1 Current notebook modules

Notebook	Important Cells / Functions	Status
CoralGuard_X_V2_FINAL_Teacher_Student_LOCAL_Windows.ipynb	Cells 1–17: environment, paths, CoralSCOP pseudo masks, SEG dataset, student training/validation, dual-view classifier, OOD, SAM2 video.	Completed experiment notebook; primary runtime superseded
CoralGuard_X_ACCURACY_FIRST_YOLO_CoralSCOP_DualView_SAM2_LOCAL.ipynb	Cells 1–17: accuracy-first config, old detector, CoralSCOP direct masks, dual-view classifier, OOD, image smoke test, video seed pass, SAM2 render.	Current main notebook
Coral_Video_V2_BoTSORT_Stable_Boxes_Large_Text.ipynb	BoT-SORT + bbox smoothing + species EMA.	Historical temporal design; concepts reused
Coral_End_to_End_Stage3_MobileNetV4_V3_Validation.ipynb	Integrated detector/classifier validation and error attribution.	Historical E2E benchmark

9.2 Critical mask normalization fix

```
def normalize_mask_2d(mask):
    mask = np.asarray(mask)
    mask = np.squeeze(mask)
    if mask.ndim == 3:
        if mask.shape[-1] in (1, 3, 4):
            mask = mask[..., 0]
        elif mask.shape[0] in (1, 3, 4):
            mask = mask[0]
    if mask.ndim != 2:
        raise ValueError(f"Expected 2D mask, got {mask.shape}")
    return (mask > 0).astype(bool)
```

Purpose: normalize CoralSCOP/PNG masks before np.where, ROI slicing, or RGB boolean indexing. This fix resolved both mask_bbox dimensionality errors and the Dual-View Cell 7 H×W×1 vs H×W×3 boolean-index failure.

9.3 Student training configuration (completed)

```
SEG_EPOCHS = 150
SEG_IMGSZ = 1024
batch = 16   # actual completed run
optimizer = AdamW
lr0 = 1e-4
lrf = 0.02
weight_decay = 5e-4
cos_lr = True
patience = 40
mosaic = 0.15
copy_paste = 0.10
multi_scale = 0.10
max_det = 500
```

The completed run took 232.61 minutes and produced best.pt and last.pt. best.pt is the model-selection checkpoint.

9.4 Dual-View speed patch

```
# Instead of two separate backbone calls:
# f_context = backbone(context)
# f_masked  = backbone(masked)

x = torch.cat([context, masked], dim=0)
features = backbone(x)
B = context.shape[0]
f_context = features[:B]
f_masked  = features[B:]
embedding = fusion(torch.cat([f_context, f_masked], dim=1))
```

Additional supplied optimizations: preload resized 300×300 dual-view JPEGs into RAM; AMP FP16; channels_last; cuDNN benchmark; TF32; fused AdamW; resume from last checkpoint. Final speedup was not yet reported in the conversation.

9.5 Current runtime configuration and quick-demo patch

```
# Accuracy-first candidate behavior
DET_CANDIDATE_CONF = 0.20
CORALSCOP_MIN_SCORE = 0.35
CORALSCOP_MIN_PIXELS = 40

# Notebook default was 3; user tested 1, then moved to 5 for speed.
REFRESH_STRIDE = 5

# Quick doctor demo recommendations (temporary):
video_paths = video_paths[:1]
KEEP_TRACKS = 25
CLS_STRIDE = 5
# FFmpeg: preset=veryfast, CRF=16
```

Not yet integrated warning
The quick-demo/top-25-track and accelerated EfficientNet cells were supplied as replacement cells after the main notebook was created. Treat them as current patch knowledge, not automatically present in the saved notebook unless manually pasted.

10. Naming & Terminology Dictionary

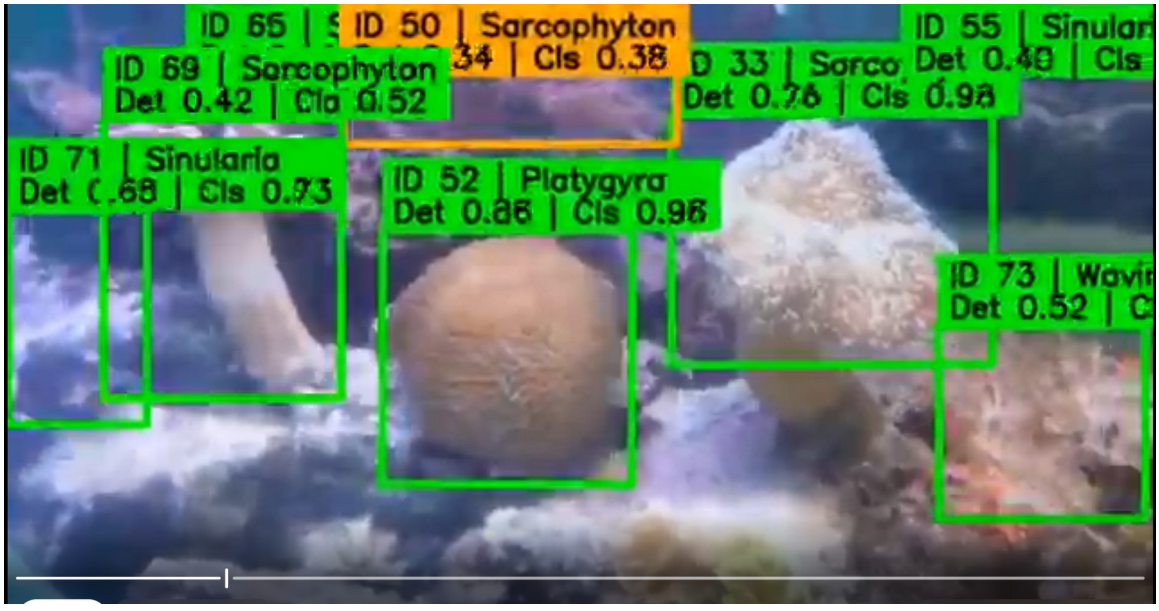
Term	Meaning	Used In
target_coral	The detector's single object class.	Detection
ROI	Region of Interest crop corresponding to one coral box/mask.	Classification
Stage 1 / Stage 2 / Stage 3	YOLO11n baseline, enhanced underwater fine-tune, targeted small/crowded fine-tune.	Historical detector R&D
Stage 4 P2	YOLO11n architecture with P2/P3/P4/P5 output strides 4/8/16/32.	Historical detector experiment
CoralSCOP	Coral-specialized box-prompt segmentation model used as mask teacher/refiner.	Segmentation
Teacher mask / pseudo mask	Mask generated by CoralSCOP from an existing GT box; not manual mask GT.	Teacher→Student
YOLO26m-SEG Student	Instance segmentation model trained from CoralSCOP pseudo masks.	Fast/backup segmenter
Dual-View	Classifier input consisting of masked foreground plus +10% context crop.	Classifier
NON_CORAL	Proxy negative class generated from conservative background crops.	Validity/classifier
Other/Unknown Coral	Pragmatic rejection label for uncertain known-class predictions.	OOD
OOD	Out-of-distribution / open-set style rejection logic; pragmatic here.	Classifier calibration
Prototype	Normalized mean embedding of a known class used for cosine similarity.	OOD calibration
EMA	Exponential moving average for species probabilities / historically box smoothing.	Video stability
Refresh frame	Frame where the full YOLO + CoralSCOP	Video

	spatial stack is rerun.	
REFRESH_STRIDE	Number of source frames between refreshes.	Runtime speed/accuracy control
Seed record	CoralSCOP mask/box/track record used to condition SAM2.	Video
SAM2.1	Temporal mask memory/propagation model used for multi-object video masks.	Video
BEST_SEG_CONF	Validation-tuned operating confidence for SEG student against original GT boxes; observed 0.40.	SEG evaluation

11. UI / UX Documentation

There is no formal graphical frontend. The current user interface is the VS Code/Jupyter notebook plus generated annotated video. UI/UX requirements therefore concern cell organization, progress reporting, configuration visibility, restart safety and video overlay readability.

Screen / Surface	User	Purpose / Actions	Key Fields	Validation / Behavior	Connected Backend
Environment / Config Cells	Research operator	Verify GPU, paths, versions; adjust batch/thresholds/strides.	Workspace paths, model paths, DET_CANDIDATE_CONF, REFRESH_STRIDE, batch sizes.	Fail fast if CUDA/path/checkpoint missing.	Local filesystem + PyTorch/Ultralytics
Training Cells	Research operator	Start/resume training and observe epoch metrics.	Epochs, batch, optimizer, augmentations.	Checkpoint best/last; early stopping.	Model output folders
Validation / Calibration Cells	Research operator	Review mAP/F1/confidence and promote checkpoints.	Validation split, thresholds.	Test isolation; schema checks.	CSV/JSON reports
Video Seed Pass	Research operator	Observe refresh progress, tracks, masks, ETA.	Video list, refresh stride.	Resume artifacts and track cap/pruning.	Frame cache + seed metadata
Final Video Overlay	Researcher / academic reviewer	See object ID, mask, species/Unknown and confidence.	SAM2 masks + classifier state.	Avoid clutter/duplicate instances.	FFmpeg MP4



Historical stable-video overlay example: tracked IDs, species labels and detector/classifier confidence. New architecture intends to replace box-only emphasis with mask overlays.

12. Roles & Permissions

No software RBAC or authentication layer exists in the project. The roles below are operational roles inferred from the workflow, not implemented application permissions.

Role	Responsibilities	Screens / Artifacts	Editable	Viewable	Approval Authority /

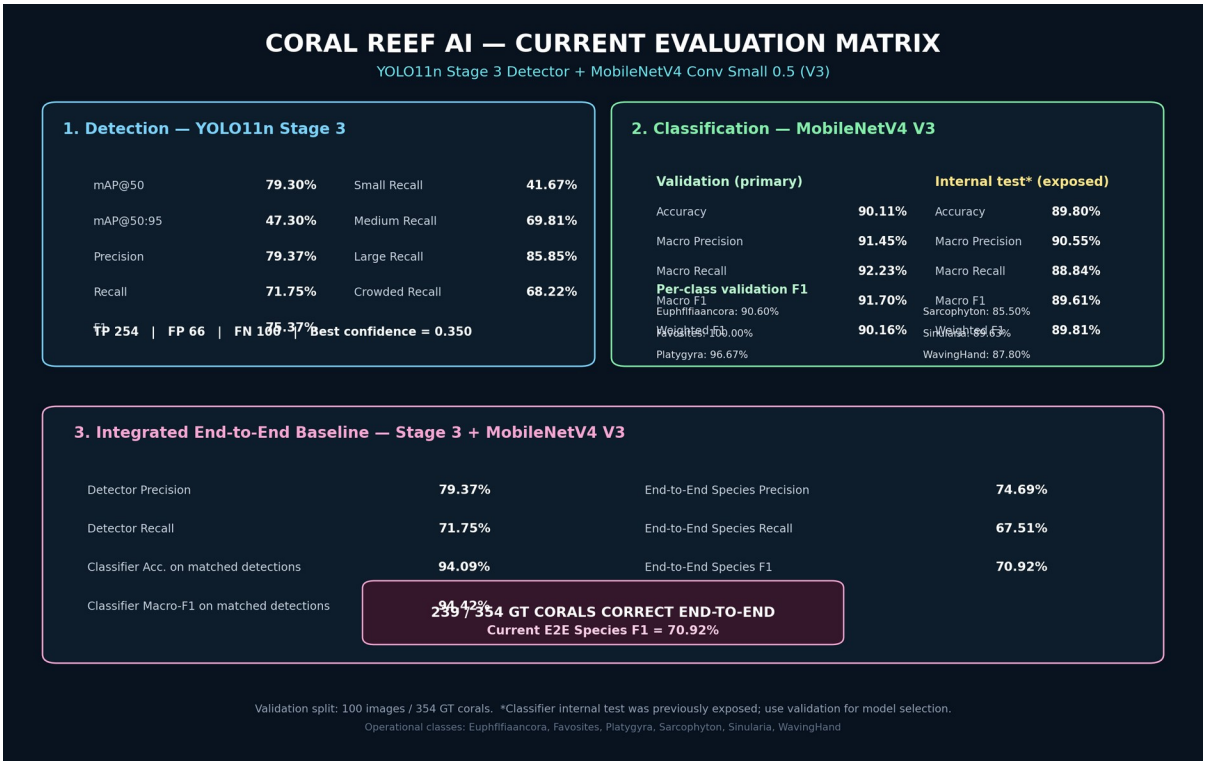
					Restrictions
Research/ Development Operator	Run notebooks; edit configs; train/evaluate; inspect outputs.	All notebooks, datasets and output folders.	Local code/configs/artifacts.	All project artifacts.	Responsible for not touching held-out test during tuning.
Academic Reviewer (“doctor”)	Review demonstration results and project progress.	Final videos, diagrams, metrics/presentations.	None implemented.	Exported deliverables.	No software approval workflow implemented.
Future Dataset/Annotation Reviewer	Manual mask/taxonomy QA would improve scientific validity.	QC images/manifests.	Not implemented.	Potential future review artifacts.	Pending role, no permissions system.

13. Automation & Triggers

Automation	Trigger	Conditions	Action	Source	Destination	Failure Behavior
AUTO-01	Notebook cell execution	Dataset/model checks pass	Generate derived manifests/crops/masks	Source files	Output folders	Fail fast exception; many stages resume-safe
AUTO-02	During model training	Validation Macro-F1 or mAP improves	Save best checkpoint; always maintain last checkpoint where supported	Training batches	weights/*.pt or .pth	Early stopping via patience
AUTO-03	SEG validation	Confidence sweep over validation predictions	Select BEST_SEG_CONF and write operating config	Original validation GT	student_operating_config.json	Does not touch test
AUTO-04	Dual-view OOD calibration	Classifier/ embedding evaluation completes	Compute prototypes and similarity thresholds	Train/validation embeddings	ood_calibration.json + NPZ	Pragmatic OOD only
AUTO-05	Video refresh frame	frame_idx % REFRESH_STRIDE == 0	YOLO candidate detection + CoralSCOP masks + association	Video frame	Seed records/masks	Dominant runtime cost
AUTO-06	Video classification frame	frame_idx % CLS_STRIDE == 0	Classify SAM2 masks and update EMA probabilities	Propagated masks	Track classifier state	Reduces redundant classifier calls
AUTO-07	Final render	SAM2 frame output available	Draw overlays and pipe BGR frames to FFmpeg	Annotated frame stream + source audio	H.264 MP4	Raise on FFmpeg failure

14. Reports / Search / Dashboards

Report / Artifact	Purpose	Source	Notes
Coral_Project_Stage3_MobileNetV4_Evaluation_Matrix.png	Historical combined detector/classifier/E2E dashboard.	Stage3 audit + MobileNetV4 V3 + integrated validation.	Key comparison artifact.
training_history*.csv	Epoch history for classifier/detector experiments.	Training loops.	Used to inspect plateau/best epoch.
teacher_mask_quality_audit.csv	Pseudo-mask pass/failure reasons and metrics.	CoralSCOP mask filtering.	Troubleshooting pseudo-label quality.
student_confidence_sweep_original_GT.csv	SEG student precision/recall/F1 across confidence.	Original validation GT.	Fair operating-point tuning for student boxes.
final_validation_classification_report.csv	Per-class classifier precision/recall/F1.	Dual-view validation.	Final classifier analysis.
final_validation_confusion_matrix.csv	Confusion matrix.	Dual-view validation.	Class error analysis.
accuracy_first_video_summary.csv	Per-video runtime/stability summary.	Final video render.	No ground-truth video accuracy claimed.
ACCURACY_FIRST_FINAL_PROJECT_SUMMARY.json	Final configuration and scientific limitations.	Final notebook summary cell.	Machine-readable handover.



Historical integrated evaluation matrix retained as the primary Stage3 + MobileNetV4 V3 baseline snapshot.

15. Calculations & Formulas

Calculation	Formula / Logic	Applied To	Notes
Intersection over Union (box)	IoU = area(intersection) / area(union)	Box matching, mask-bbox alignment, duplicate resolution.	Guard denominator with epsilon.
F1	2 × Precision × Recall / (Precision + Recall)	Detector operating point and species E2E.	Do not confuse correct-count fraction with F1.
Mask IoU	pixels(A ∩ B) / pixels(A ∪ B)	Duplicate suppression and track association.	Masks normalized to 2D boolean first.
Mask containment	pixels(A ∩ B) / min(pixels(A), pixels(B))	Detect nearly contained duplicate masks.	Current duplicate threshold around 0.85.
+10% context	Expand box width/height by 10% on each dimension and clip to image.	Classifier context view.	Preserves local environment without full-scene contamination.
CoralSCOP candidate mask quality	Weighted combination of teacher score, box alignment and reasonable area behavior.	Choose one mask from multimask output.	Exact weighting varied by notebook version; current generated notebook used 0.55/0.30/0.15.
OOD cosine similarity	normalized embedding dot class prototype.	Known-class validity gate.	Per-class threshold derived from known validation embeddings.
Species probability EMA	p_smooth = α·p_current + (1−α)·p_previous	Video label stability.	Current α = 0.25.
Refresh count	ceil(number_of_frames / REFRESH_STRIDE)	Seed-pass workload estimate.	1821 frames at stride 5 ≈ 365 refreshes.
Track association score	Weighted mask IoU + box IoU + center proximity.	Refresh-frame ID association.	Current notebook family uses mask/box/center terms; fragmentation remains pending.

16. Testing History & Regression Checklist

Test ID	Scenario	Input	Expected	Actual	Fix / Consequence	Status
TEST-001	Raw audit	646 images + 646 JSON; 2,199 objects before dedupe.	Detect malformed/missing pairs.	Valid; one duplicate found.	Duplicate removal + canonicalization.	Pass
TEST-002	Split audit	452/100/94 images and 1,540/354/304 boxes.	Leakage-safe image-level split.	Expected counts achieved.	Locked split.	Pass
TEST-003	RF-DETR baseline validation	10 epochs.	Strong reference detector.	Epoch10 mAP50:95 .5231; best regular .5373.	Retained benchmark.	Pass
TEST-004	YOLO11 Stage3 validation audit	conf ~.35.	Small/crowded improvement.	F1 .7537; small recall .4167; crowded .6822.	Retained baseline.	Pass

TEST-005	SAHI validation	Stage3 + SAHI640.	Improve small/crowded inference.	Failed badly relative to standard path.	Rejected SAHI.	Fail / rejected
TEST-006	MobileNetV4 V3 validation/test	288px + context + hard mining.	Classifier generalization.	Test Acc .8980; MacroF1 .8961.	Historical classifier final.	Pass; test historically exposed
TEST-007	Integrated Stage3 + MobileNetV4 V3	100 val images / 354 GT.	End-to-end species performance.	E2E Species F1 70.92%.	Failure attribution drove detector focus.	Pass
TEST-008	YOLO26m + EfficientNetV2-S integrated	100 val images / 354 GT.	Stronger pair.	E2E Species F1 .708978.	Pre-seg benchmark.	Pass
TEST-009	SEG student pseudo-mask validation	100 images / 352 instances.	Teacher agreement.	Mask mAP50:95 .590519; box mAP50:95 .641117.	Useful segmenter.	Pass with pseudo-GT caveat
TEST-010	SEG student original-GT confidence sweep	Original 354 GT boxes.	Fair detection operating point.	conf .40; P .7857; R .6836; F1 .7311.	Old YOLO26m kept as primary detector.	Pass / decision
TEST-011	Dual-view first epoch	Epoch 1/150.	Check classifier starts correctly.	Acc .8921; MacroF1 .8699.	Training later optimized for speed.	Partial; final best not reported
TEST-012	Accuracy-first video seed pass	5 videos; first 1821 frames.	Runtime/track behavior.	At stride 5 ~1.2–1.4 s/refresh later; tracks grew to ~67.	Speed improved; track fragmentation pending.	In progress

16.1 Regression testing checklist

- Environment: correct .venv, CUDA available, RTX A6000 visible, package versions printed.
- Paths: detector, classifier, CoralSCOP, SAM2, manifests and video ZIP/folder resolve without fallback surprises.
- Dataset counts: 452/100/94 images and 1,540/354/304 objects remain unchanged.
- ROI mapping: 1,894 train+validation positive ROIs map to CoralSCOP masks; no missing class/split.
- Mask robustness: all loaded masks normalize to H×W boolean; resized with nearest-neighbor only.
- Dual-view outputs: context/masked files exist, no duplicate manifest rows, all seven labels represented as expected.
- Classifier: checkpoint class_to_idx exactly matches runtime mapping; validation Macro-F1 reported before promotion.
- OOD: calibration files correspond to the same classifier checkpoint and class order.
- Detector: candidate threshold change is recorded; do not silently reuse old .375 threshold for the new architecture.
- Video: first smoke-test image is visually inspected for masks/duplicates before long batch run.
- Tracking: monitor unique/active track count; fail or prune if IDs grow implausibly.
- Final video: verify frame count, source FPS, audio preservation (if expected), readable overlay and no encoding artifacts.
- Scientific: held-out detector test remains untouched until final architecture is frozen.

17. What We Learned Building This System

Domain	Project-specific lesson
Architecture	A box-only pipeline can score well yet fail visually when irregular overlapping objects contaminate ROI crops. Real deployment screenshots exposed this earlier than aggregate mAP did.
Evaluation	Pseudo-label validation and original human box validation answer different questions. High teacher agreement cannot be marketed as independent segmentation accuracy.
Component responsibility	The best segmenter does not need to be the best detector. Keeping old YOLO26m for candidates and CoralSCOP for masks is more defensible than forcing YOLO26m-SEG to own both jobs.
Data engineering	Image-level split lineage must flow into every derived ROI/mask dataset; otherwise a clean detector split can become a leaky classifier split.
Small/crowded objects	Target selection logic matters. Random slicing that does not guarantee the intended small object can waste training volume.
Model experimentation	A theoretically attractive method (SAHI or P2) should not survive solely because it matches the problem description; measured behavior decides promotion.
Classifier design	Coral texture/morphology is sensitive to destructive augmentations. Context helped, but blur/downscale/heavy JPEG/noise were intentionally avoided in the stronger classifier path.
OOD	Softmax confidence alone is insufficient for unknown/background rejection. Combining NON_CORAL, confidence, margin and embedding similarity is more practical, while still requiring honest open-set caveats.
Notebook engineering	Restart-safe, schema-consistent cells are critical. Missing imports,

	implicit globals and inconsistent error-branch dictionaries caused avoidable failures.
Mask handling	Always normalize mask dimensionality immediately after reading/inference. H×W, H×W×1 and singleton-channel tensor forms must be handled explicitly.
Performance	The dominant bottleneck can move: disk I/O → duplicate backbone forwards → CoralSCOP refresh cost → SAM2 track count. Optimize measured bottlenecks one layer at a time.
Video systems	Tracking stability and runtime are architectural, not cosmetic. Refresh cadence, track association, ID pruning and classifier cadence materially determine whether the final system is usable.
Communication	The project needs both Accuracy Mode and Demo/Fast Mode. A research presentation may prioritize obtaining one high-quality representative video quickly instead of processing an entire batch.

18. Current State Snapshot

Category	Current State
Working	Canonical detection/classification dataset and split; historical detector/classifier baselines; CoralSCOP pseudo-mask generation; YOLO26m-SEG student training/validation; accuracy-first YOLO→CoralSCOP image path; dual-view dataset build; local RTX A6000 environment; video frame extraction and CoralSCOP seed-pass logic.
Partially Working / In Progress	Dual-View EfficientNetV2-S final training metrics are not fully reported; accelerated replacement cell was supplied. Video seed pass is running with stride tuning. SAM2 final end-to-end video output has not yet been confirmed in conversation.
Pending	Track fragmentation fix/pruning; first quick doctor-demo video; final SAM2 render verification; final E2E metric suite on frozen architecture; held-out detector test; explicit unknown/non-coral benchmark; human mask QA if independent segmentation metrics are desired.
Known Issues	Track IDs can grow to ~67–69 on sample video; CoralSCOP refresh remains expensive; NON_CORAL negatives may contain unlabeled coral if annotations are not exhaustive; Unknown is pragmatic only; taxonomy remains partly ambiguous.
Removed / Deprecated	Original “no segmentation” constraint; SAHI inference; YOLO11 P2 Stage4 as final detector; SEG student as primary detector/segmenter; always-run CoralSCOP on every frame (stride=1) for practical demo runtime.
Current Final Architecture	YOLO26m old detector → CoralSCOP direct mask → overlap/duplicate resolver → masked + context Dual-View EfficientNetV2-S → OOD/validity gate → SAM2.1 temporal memory → H.264 video.

18.1 Current local environment / paths

Item	Value
Workspace	D:\Coral_Local_Workspace
Python	3.12.0 in .venv
PyTorch	2.11.0+cu128
CUDA runtime	12.8
GPU	NVIDIA RTX A6000, ~48 GB VRAM
Ultralytics	8.4.117
timm	1.0.22
OpenCV	5.0.0 recorded
FFmpeg	C:\Users\sayed.abdo\AppData\Local\Microsoft\WinGet\Packages\Gyan.FFmpeg_Microsoft.Winget.Source_8wekyb3d8bbwe\ffmpeg-9.0-full_build\bin\ffmpeg.EXE
Detection data	D:\Coral_Local_Workspace\data\Coral_Reef_AI_Project\01_Data\Final\Detection_v1
ROI manifest	D:\Coral_Local_Workspace\data\Coral_Reef_AI_Project\04_Dataset_Splits\classification_roi_manifest_split.csv
Current final notebook	CoralGuard_X_ACCURACY_FIRST_YOLO_CoralSCOP_DualView_SAM2_LOCAL.ipynb
Expected final video root	D:\Coral_Local_Workspace\outputs\CoralGuard_X_ACCURACY_FIRST\06_Final_Videos

18.2 Immediate next steps

1. Finish / confirm one Dual-View classifier best checkpoint and record its final validation Macro-F1, per-class report and confusion matrix.
2. Use the already completed first-video seed metadata if available; do not rerun all five videos for the urgent academic demo.
3. For quick demo: limit to first video, retain top ~25 stable seed tracks, classify every ~5 frames, encode with FFmpeg veryfast/CRF16.
4. Inspect the first SAM2 output visually for ID fragmentation, duplicate masks and label stability before processing more videos.
5. Then improve refresh-frame association / stale-track pruning so the active ID count reflects actual simultaneous colonies rather than accumulated historical IDs.
6. Freeze architecture and thresholds; only then use held-out detector test and report final E2E metrics without further tuning.

19. Missing / Unclear Information

Gap ID	Missing / Unclear Item	What is known
GAP-001	Final Dual-View EfficientNetV2-S best epoch/metrics after the speed-optimized cell.	The conversation only explicitly shows Epoch 1: Acc .8921, MacroF1 .8699 before later optimization.
GAP-002	Whether the final accuracy-first SAM2 MP4 was successfully generated.	The conversation currently shows seed-pass progress, not final output verification.
GAP-003	Independent mask ground truth.	No human instance-mask GT is available; pseudo-mask metrics are teacher agreement.
GAP-004	True open-set test data.	No explicit unseen coral species benchmark; Unknown remains rejection logic.
GAP-005	Formal labels for dead coral/algae.	Not present in the audited six-label dataset.
GAP-006	Exhaustiveness of coral annotations for background-negative generation.	NON_CORAL may be contaminated by unlabeled coral if annotation coverage is incomplete.
GAP-007	Final track association thresholds after fragmentation optimization.	Current association produces many IDs on example videos; final pruning logic is pending.
GAP-008	CoralSCOP commercial licensing clearance.	Conversation recorded CC BY-NC-SA 4.0/non-commercial restrictions; verify legal terms before any commercial product use.
GAP-009	Final real-time FPS.	Never established for the final accuracy-first pipeline; do not invent.
GAP-010	Final held-out test performance for frozen architecture.	Intentionally pending until model/threshold selection is complete.

20. AI Handover Context

The following block is designed to be copied into a new ChatGPT conversation as a self-contained project handover. It intentionally prioritizes current decisions plus critical historical constraints.

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You are continuing a Computer Vision project called CoralGuard-X / Coral Reef AI. Treat this context as the
project source of truth unless I explicitly provide newer results.

PROJECT PURPOSE
Build an underwater coral system that detects coral instances, isolates them accurately enough for species
classification, rejects uncertain/non-coral candidates pragmatically, and produces stable annotated video.
Original scope was Detection → Crop → Classification → Video → Real-Time with no segmentation, but real-
video failures forced an instance-segmentation pivot.

DATA
Canonical detection dataset: 646 images, 2,198 final GT boxes after removing one duplicate. Image-level
split: train 452 images/1,540 boxes; validation 100/354; test 94/304. Detector is single-class
target_coral. Classification ROIs use the same lineage: 1,540/354/304. Six operational species labels:
Euphflfiaancora, Favosites, Platygyra, Sarcophyton, Sinularia, WavingHand. Taxonomy is partly ambiguous; do
not silently “correct” labels. Held-out detector test must not be used for tuning. The old classifier test
was historically exposed and must not be reused for model selection.

HISTORICAL BASELINES
RF-DETR Medium 10 epoch final validation: mAP50:95 .5231, mAP50 .8271, mAP75 .5767, mAR@500 .7288, F1
.7808, P .8030, R .7599; best regular mAP ~.537313.
YOLO11n Stage3 standard audit at conf .35: P .7937, R .7175, F1 .7537, small recall .4167, medium .6981,
large .8585, crowded .6822; mAP50 ~.793, mAP50:95 ~.473. SAHI was rejected. Stage4 P2 was under-converged
and not promoted.
MobileNetV4 V3: 288px +10% context + train-only hard mining; test Acc .8980, MacroF1 .8961, Sarcophyton
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test F1 .8539.
Stage3+MobileNet V3 integrated validation: 100 images/354 GT; 254 matched detections; 239 correct detection+species; E2E Species P .7469, R .6751, F1 .7092. Failure attribution: 68 detector misses, 32 localization errors, 15 classification errors.
Stronger pre-seg pair: YOLO26m validation mAP50:95 .507131, mAP50 .806921, P .836268, R .714689. EfficientNetV2-S validation Acc .889831, MacroF1 .883817. Integrated E2E F1 .708978 at detector conf .375.

SEGMENTATION PIVOT
Real video showed overlapping boxes, excess rock/background inside ROIs, uncertain/Unknown predictions and dead-looking/algae-like structures. Instance segmentation was introduced. CoralSCOP is used as a coral-specific box-prompt teacher/refiner. There is no explicit Dead Coral or Algae class.

TEACHER→STUDENT EXPERIMENT
Existing GT boxes → CoralSCOP pseudo instance masks → mask quality filter → YOLO26m-SEG student. Student trained 150 epochs at imgsz1024; completed in 232.61 minutes. Pseudo-mask validation: box mAP50:95 .641117, box mAP50 .811005, box mAP75 .687784, mask mAP50:95 .590519, mask mAP50 .787498, mask mAP75 .630761. On original validation GT boxes, best confidence .40: TP242 FP66 FN112, P .785714, R .683616, F1 .731118. Therefore the SEG student is useful as fast/backup segmentation but is NOT the primary detector in the accuracy-first architecture.

CURRENT FINAL ARCHITECTURE
RAW FRAME → OLD YOLO26m DETECTOR → candidate boxes → CoralSCOP directly on each candidate → precise instance masks → mask QC/overlap/duplicate resolver → masked coral view +10% context view → Dual-View EfficientNetV2-S → known species / Other-Unknown Coral / Non-Coral validity logic → SAM2.1 temporal mask memory → stable ID+mask+species annotated H.264 video.
Use old YOLO26m as the primary detector because its original-GT operating behavior was stronger than the SEG student. CoralSCOP is primary accuracy mask source. YOLO26m-SEG student is a fast/backup mode.

CURRENT RUNTIME / OOD RULES
Accuracy-first candidate confidence = .20. CoralSCOP min score ~.35, min pixels 40. Duplicate mask IoU ~.65, containment ~.85. Dual-view context padding 10%. Classifier has six known species + NON_CORAL proxy class. OOD thresholds in design: known confidence .50, top1-top2 margin .08, NON_CORAL .45, plus per-class prototype cosine similarity. Unknown is pragmatic rejection only because no true unseen-species validation set exists. SAM2 model: facebook/sam2.1-hiera-large. Species EMA alpha .25.

VIDEO SPEED STATE
Running CoralSCOP every frame (REFRESH_STRIDE=1) was too slow (~1.5–2 s per refresh). The user moved to REFRESH_STRIDE=5; first video has 1821 frames → about 365 refreshes and later speed around ~1.2–1.4 s/refresh. Track IDs grew to ~67–69, indicating fragmentation; track pruning/stronger association is pending. Immediate priority is one video quickly for an academic doctor. Quick-demo recommendation: process only first video, reuse completed seed metadata, keep top ~25 stable tracks, CLS_STRIDE=5, FFmpeg preset veryfast / CRF16. Do not rerun all 5 videos unnecessarily.

LOCAL ENVIRONMENT
Windows workspace D:\Coral_Local_Workspace. Python 3.12.0, torch 2.11.0+cu128, CUDA 12.8, NVIDIA RTX A6000 ~48GB, ultralytics 8.4.117, timm 1.0.22. Main detection data: D:\Coral_Local_Workspace\data\Coral_Reef_AI_Project\01_Data\Final\Detection_v1. ROI manifest: ...\04_Dataset_Splits\classification_roi_manifest_split.csv. CoralSCOP checkpoint: D:\Coral_Local_Workspace\models\vit_b_coralscop.pth. Main current notebook: CoralGuard_X_ACCURACY_FIRST_YOLO_CoralSCOP_DualView_SAM2_LOCAL.ipynb. Expected final video root: D:\Coral_Local_Workspace\outputs\CoralGuard_X_ACCURACY_FIRST\06_Final_Videos.

CRITICAL TROUBLESHOOTING
- Normalize every mask to strict H×W boolean before np.where/indexing; this fixed both mask_bbox and H×W×1 boolean-index errors.
- Cell 6 student mask audit previously had KeyError 'score'; every branch must return the same audit schema.
- After restart, reload checkpoint/config/manifest state; do not retrain completed models.
- Dual-view data build was accelerated by grouping ROIs by source image and skipping existing outputs.
- Dual-view classifier speed patch: cache resized 300×300 images in RAM; single concatenated backbone forward for context+masked; AMP; channels_last; TF32; fused AdamW; resume checkpoint.
- Do not interpret pseudo-mask mAP as human segmentation GT accuracy.
- Do not claim final FPS, flicker or final video accuracy unless actually measured.

CURRENT STATUS / NEXT STEPS
Final dual-view best metrics after the speed-optimized training cell are not yet recorded in the conversation. Accuracy-first video seed pass is in progress; final SAM2 MP4 is not yet confirmed. Next: obtain first doctor-demo video, inspect track fragmentation/duplicates, then improve association/pruning, freeze architecture, and only then run held-out final evaluation.

Appendix A. Metrics History

Model / Stage	Metric	Value	Interpretation
RF-DETR Medium, epoch 10	Val mAP50:95	0.5231	Historical strong baseline
RF-DETR Medium, best regular	Val mAP50:95	0.537313	Epoch 8 recorded
RF-DETR Medium, epoch 10	mAP50 / mAP75	0.8271 / 0.5767	P .8030 / R .7599 / F1 .7808
YOLO11n 10ep	mAP50:95 / mAP50	0.436 / 0.697	P .719 / R .593
YOLO11n Stage2	mAP50:95 / mAP50	0.4439 / 0.7727	P .7576 / R .6864
YOLO11n Stage3	mAP50:95 / mAP50	~0.473 / ~0.793	Custom F1 .7537 @ conf .35
YOLO11n Stage3	small / crowded recall	0.4167 / 0.6822	Main targeted groups
YOLO11n P2 Stage4	mAP50:95 / mAP50	0.265 / 0.596	Under-converged; not promoted
MobileNetV4 V3	Test Accuracy / MacroF1	0.8980 / 0.8961	Historical test exposed
Stage3 + MobileNet V3	E2E Species F1	0.7092	239/354 correct E2E; do not confuse .6751 recall with F1
YOLO26m	Val mAP50:95 / mAP50	0.507131 / 0.806921	P .836268 / R .714689
EfficientNetV2-S	Val Acc / MacroF1	0.889831 / 0.883817	Pre-seg single-view strong model
YOLO26m + EffNetV2-S	E2E Species F1	0.708978	conf .375; 229 correct detection+species
YOLO26m-SEG Student	Pseudo box mAP50:95	0.641117	Teacher-mask dataset
YOLO26m-SEG Student	Pseudo mask mAP50:95	0.590519	Teacher agreement, not human mask GT
YOLO26m-SEG Student	Original-GT P/R/F1	0.785714 / 0.683616 / 0.731118	Best conf .40; TP242 FP66 FN112
Dual-View EffNetV2-S	Epoch 1 Acc / MacroF1	0.8921 / 0.8699	Only explicit current training epoch result recorded

Appendix B. Project Artifact Inventory

Artifact	Purpose
YOLO11n_Coral_Detection_10Epochs.ipynb	YOLO11n detection baseline.
YOLO11n_Validation_Error_Audit.ipynb	Validation error/size/crowding audit.
YOLO11n_Stage3_Targeted_Sliced_Finetune_SAHI.ipynb	Stage3 targeted training + rejected SAHI evaluation.
YOLO11n_Stage3_1_SmallHardPriority_Localization_Finetune.ipynb	Stage3.1 small/hard-priority localization fine-tuning.
YOLO11n_P2_Stage4_SmallPriority_Coral.ipynb	P2 small-priority experiment.
YOLO11n_P2_Stage4_1_HybridRecovery_Standalone.ipynb	Stage4 recovery experiment notebook.
Coral_Classification_MobileNetV4_Small050.ipynb	MobileNetV4 classification pipeline.
Coral_End_to_End_Stage3_MobileNetV4_V3_Validation.ipynb	Historical E2E integrated evaluation.
Coral_Video_Inference_Stage3_MobileNetV4_V3.ipynb	Initial video inference.
Coral_Video_V2_BoTSORT_Stable_Boxes_Large_Text.ipynb	Tracking/stabilization video version.
RFDETR_Large_ConvNeXtSmall_Full_Coral_Pipeline.ipynb	Alternative RF-DETR/ConvNeXt full pipeline.
YOLO26m_EfficientNetV2S_Full_Coral_Pipeline.ipynb	Stronger pre-seg YOLO26m + EfficientNetV2-S pipeline.
YOLO26m_Continue_50ep_From_Best_Colab.ipynb	Continuation experiment.
YOLO26m_EfficientNetV2S_PLUS150_EACH_Colab.ipynb	Planned +150 each continuation notebook.
YOLO26m_EfficientNetV2S_Colab_Resume_VIDEO_ONLY.ipynb	Video-only resume path.
CoralGuard_X_FINAL_Hybrid_Coral_Pipeline_LOCAL_Windows.ipynb	Initial CoralSCOP inference-refinement hybrid.
CoralGuard_X_V2_FINAL_Teacher_Student_LOCAL_Windows.ipynb	Teacher→Student segmentation architecture.
CoralGuard_X_ACCURACY_FIRST_YOLO_CoralSCOP_DualView_SAM2_LOCAL.ipynb	Current final accuracy-first notebook.
Coral_Project_Stage3_MobileNetV4_Evaluation_Matrix.png	Historical metric dashboard.
plan full.png	Early project plan / roadmap visual.
coral_video.zip	Video test input archive.

Excluded unrelated material
A RECOVERY-X2 log containing B4_RMSE, SAM, PSNR, SSIM and DC was identified as unrelated to the coral project and is intentionally not merged into coral metrics.

Appendix C. Mermaid Reference Diagrams

C.1 Teacher→Student training

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flowchart TD
    A[Existing GT Boxes] --> B[CoralSCOP Teacher]
    B --> C[Pseudo Instance Masks]
    C --> D[Mask Quality Filter]
    D --> E[YOLO Polygon Segmentation Labels]
    F[YOLO26m-seg Base] --> G[YOLO26m-SEG Student]
    H[Old YOLO26m compatible weights] --> G
    E --> G
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    G --> I[Pseudo-mask Validation]
    G --> J[Original-GT Box Confidence Sweep]
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C.2 Video runtime

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flowchart LR
    V[Video Frames] --> K[Refresh Frame]
    K --> Y[YOLO26m Detector]
    Y --> C[CoralSCOP Masks]
    C --> S[SAM2.1 Conditioning]
    S --> P[Mask Propagation]
    P --> Q{Classification frame?}
    Q -- yes --> E[Dual-View EfficientNetV2-S + EMA]
    Q -- no --> R[Reuse smoothed state]
    E --> O[Overlay / H.264]
    R --> O
    P --> K
```



Early project plan visual preserved as historical planning evidence.

End of document.

This reconstruction preserves current decisions, superseded designs, measured metrics, troubleshooting fixes, known gaps and next actions as of the last project message available on 03 September 2026.